Lecture 12: Bayesian Statistics

IEOR E4709: Statistic Analysis and Time Series
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1 Bayesian Statistics

In our first class on probability, we all learned the Bayes formula. Let us briefly restate it:

$$\mathbb{P}\left(\mathcal{H}|\mathcal{D}\right) = \frac{\mathbb{P}\left(\mathcal{D}|\mathcal{H}\right)\mathbb{P}\left(\mathcal{H}\right)}{\mathbb{P}\left(\mathcal{D}\right)}$$

The prior $\mathbb{P}(H)$ is the probability that the hypothesis H is true before the data is considered. The posterior $\mathbb{P}\left(\mathcal{H}|\mathcal{D}\right)$ is the probability that H is true conditional on observing the data $\mathcal{D}$. The probability of observing the data $\mathcal{D}$ given that H is true is $\mathbb{P}\left(\mathcal{D}|\mathcal{H}\right)$. $\mathbb{P}\left(\mathcal{D}\right)$ is the total probability of the data taking into account all possible hypotheses. If the prior and likelihood are known for all hypotheses, then Bayes formula computes the posterior exactly.

In most experiments, the prior probabilities on hypotheses are not known. In this case, we either postulate a prior (Bayesian approach) or do our best using only the likelihood (frequentist approach).

Next, let us write the Bayes rule in terms of probability density functions. Let θ be unknown parameter, which we treat as a random variable and we put a probability distribution on it. This probability is then updated based on the received data. More specifically, let $f(\theta)$ be the *prior* probability density function on θ. We are also given the *conditional probability density* function, that is $f(x|\theta)$, which is the probability of observing the data x conditioning on the parameter θ. The *Bayesian posterior* belief of θ after observing the data x is the conditional probability density of θ given the sample, and can be computed by a direct application of Bayes' rule

$$f\left(\theta|x\right) = \frac{f(x|\theta)f(\theta)}{\int f(x|\theta')f(\theta')d\theta'}$$

Example: Let $X_1, X_2, \ldots, X_{90}$ stand for indicators of economic shocks in the past 90 trading days. An economic shock is an unexpected event that produces a significant change within an economy. Examples include shocks in the supply of commodities, rapid devaluation of a currency producing a shock to the import/export industry, or unpredictable political events (e.g. Trumps' election). Suppose no event occurs, i.e, $X_i = 0$, for $i = 1, \ldots, 90$. A frequentist statistician would estimate that the probability p of observing a shock is zero. Consider a Bayesian, who puts a beta distribution on the parameter p, given by $f(p) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}p^{\alpha-1}(1-p)^{\beta-1}$. (if $\alpha = \beta = 1$, the Beta distribution becomes a $(0,1)$ uniform distribution and $f(p) = 1$ for any $p \in [0,1]$). The conditional

probability density is given by

$$f(X|p) = \prod_{i=1}^{n} p^{X_i}(1-p)^{1-X_i} = p^{n\bar{X}}(1-p)^{n(1-\bar{X})}$$

We can apply Bayes rule to obtain the posterior distribution of p given the data X. Applying Bayes's rule, we obtain

$$f(p|X) \propto f(X|p)f(p) \propto p^{n\bar{X}+\alpha-1}(1-p)^{n(1-\bar{X})+\beta-1}.$$

Hence, the posterior distribution of the parameter p is $Beta(\alpha+n\bar{X}, \beta+n(1-\bar{X}))$. It is well known that the mean of a β distribution with (α, β) parameters is $\frac{1}{1+\frac{\beta}{\alpha}}$. Hence, the mean of our posterior distribution is

$$\frac{\alpha+n\bar{X}}{\alpha+\beta+n} = \frac{\alpha+\beta}{\alpha+\beta+n} \times \frac{\alpha}{\alpha+\beta} + \frac{n}{\alpha+\beta+n}\bar{X} \tag{1}$$

The above expression admits an intuitive interpretation. It is a convex combination of the prior mean $\frac{\alpha}{\alpha+\beta}$, and the sample mean $\bar{X}$. Assume that the Bayesian uses a totally uninformative prior, that is the uniform distribution $f(p) = 1$. Using Eq. (1), the mean of the posterior distribution equals $\frac{1+1}{1+1+90} \times \frac{1}{1+1} + \frac{90}{1+1+90} \times 0 = 0.0109$. Which one do you find more reasonable, the Bayesian or the frequentist estimate of the probability of no shocks to the economy?

1.1 Conjugate Prior

Bayesian statistics is often used in a sequential procedure. One starts from a prior, then combines it with the likelihood of the data, to obtain a posterior. This posterior would then be viewed as the new prior, combined with the likelihood of new data, to obtain another posterior. To make this sequential procedure computationally tractable, it is desirable that the posterior probability belongs to the same parametric class of probability distributions as the prior. This is achieved through the concept of conjugate prior, which we review next.

A family of distributions is **conjugate** with respect to a likelihood if, for any prior in this family, the posterior is in the same family. Conjugate prior and likelihood facilitate the computation of a posterior.

Example 1: Assume a Bernoulli random variable X with parameter p. Assume that the prior distribution on p is $f(p) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}p^{\alpha-1}(1-p)^{\beta-1}$. The likelihood function is $f(x|p) = p^x(1-p)^{1-x}$. Thus the posterior is $f(p|x) \propto f(x|p)f(p) \propto p^{\alpha+x-1}(1-p)^{\beta-x}$. We can thus conclude that the posterior distribution is $Beta_{(\alpha+x,\beta+1-x)}$. So, the beta distribution is conjugate with respect to the Bernoulli distribution.

Example 2: Consider a Gaussian distribution, where $X_1, X_2, \ldots, X_n$ are i.i.d. normal samples with unknown mean μ and known variance σ^2. Suppose the prior of μ follows a $Gaussian(a, b^2)$. The likelihood function is

$$f(x|\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}},$$

where $x = (x_1, x_2, \ldots, x_n)$. The posterior distribution is

$$f(\mu|x) \propto f(x|\mu)f(\mu) \propto e^{-\frac{1}{2\sigma_n^2}(\mu-\mu_n)^2},$$

where $\sigma_n^2 := \frac{b^2\sigma^2}{\sigma^2+nb^2}$, $\mu_n := \frac{\sigma^2}{\sigma^2+nb^2} \times a + \frac{nb^2}{\sigma^2+nb^2} \times \bar{X}$. Hence the posterior is $Gaussian(\mu_n, \sigma_n^2)$.

Question: What is a 95% Bayesian confidence interval for the mean? It is $[\mu_n - z_{97.5\%}\sigma_n, \mu_n + z_{97.5\%}\sigma_n]$, where $z_{97.5\%}$ is the 97.5% quantile of the standard normal distribution.

Bayesian hypothesis testing. We use the posterior probabilities of the null and alternative hypothesis. More specifically, consider the null hypothesis $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \notin \Theta_0$. The Bayesian computes $\mathbb{P}(\theta \in \Theta_0|X)$, and $\mathbb{P}(\theta \notin \Theta_0|X)$, and rejects H_0 in favor of H_1 if the posterior $\mathbb{P}(\theta \notin \Theta_0|X) > \mathbb{P}(\theta \in \Theta_0|X)$.

1.2 How to choose priors?

One possibility is to choose a prior which conjugates to the likelihood. Another possibility is to choose subjective priors. A third possibility is to estimate priors from data, following an empirical Bayesian approach.

If you have no knowledge of the parameter a priori, a fourth possibility is to choose a non-informative prior. For instance, assume $X \sim Bin(n, p)$. A plausible noninformative prior on the parameter p would be $p \in U[0, 1]$, i.e., uniform on the $[0, 1]$ interval. However, by doing so, we would obtain that $\sqrt{p}$ has a higher density near one than near 0. This is because the square root of a number in $(0, 1)$ is larger than the number itself, thus we know intuitively that the distribution of $\sqrt{p}$ must concentrate its probability toward the right end of the interval $(0, 1)$. (Specifically, the density function would be $f_{\sqrt{p}}(x) = 2x$ for $0 < x < 1$). Thus *ignorance* about p seems to lead to *knowledge* about $\sqrt{p}$, which is paradoxical. We would like the prior to be invariant under reparametrization. We will achieve this by using the Fisher information.

Recall that the Fisher information equals $I(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial \log f(X|\theta)}{\partial \theta} \right)^2 \right]$. Let us define the *Jeffreys prior* as $\pi(\theta) \propto I(\theta)^{1/2}$. Consider a model $X \sim f(x|\theta)$, and its reparametrized version $X \sim g(x|\eta)$, where $\eta = h(\theta)$ and h is a differentiable monotone transformation of θ. Applying the chain rule, we would have that

$$\begin{aligned} I(\theta) &= \int \left(\frac{\partial}{\partial \theta} \log f(x|\theta) \right)^2 f(x|\theta) dx \\ &= \int \left(\frac{\partial}{\partial \theta} \log g(x|\eta) \right)^2 g(x|\eta) dx \\ &= \int \left(\frac{\partial}{\partial \eta} \log g(x|\eta) \Big|_{\eta=h(\theta)} h'(\theta) \right)^2 g(x|h(\theta)) dx \\ &= (h'(\theta))^2 I(h(\theta)). \end{aligned}$$

Next, recall that if X is a r.v. with pdf $f_X(x)$ that is continuous on its support, and $Y = g(X)$ where g is a monotone function of X, then the pdf f_Y of Y is given by

$$f_Y(y) = f_X(g^{-1}(y)) | \frac{\partial}{\partial y} g^{-1}(y) |$$

In our case $y = h(\theta)$, hence

$$\pi_Y(y) = \pi_x(h^{-1}(y)) \big| \frac{d}{dy} h^{-1}(y) \big|,$$

thus

$$\pi(h(\theta)) = \pi(\theta)\Big|\frac{d}{dh(\theta)}h^{-1}(h(\theta))\Big| = \pi(\theta)\left|\frac{d\theta}{dh(\theta)}\right| = \frac{\pi(\theta)}{|h'(\theta)|}.$$

Hence, for the Jeffreys prior $\pi(\theta)$, we have

$$\pi(h(\theta)) = \pi(\theta)\left|h'(\theta)\right|^{-1} \propto I(\theta)^{1/2}\left|h'(\theta)\right|^{-1} = I(h(\theta))^{1/2}$$

and thus the prior is invariant under reparametrization. The Jeffreys prior favors values of θ for which $I(\theta)$ is large. Hence, it minimizes the effect of the prior distribution relative to the information in the data.

Example: Let $X \sim Bin(n, p)$, where n is known, so that $f(x|p) = \binom{n}{x}p^x(1-p)^{n-x}$. Then

$$\frac{\partial^2 \log f(x|p)}{\partial^2 p} = -\frac{x}{p^2} - \frac{n-x}{(1-p)^2}.$$

Taking expectation and multiplying by -1 (we are using the equivalent representation $I(p) = -\mathbb{E}_p\left[\frac{\partial^2 \log f(X|p)}{\partial^2 p}\right]$, we obtain: $I(p) = \frac{n}{p} + \frac{n}{1-p} = \frac{n}{p(1-p)}$. Thus the Jeffreys prior is

$$\pi(p) \propto (p(1-p))^{-1/2},$$

which we recognize as the $Beta_{(1/2,1/2)}$ distribution.